Behavioral Biometric-Based Continuous Authentication System for Secure Mobile Banking Using Machine Learning

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Abstract

The rapid growth of mobile banking has increased the demand for secure and seamless user authentication. Conventional methods such as passwords, PINs, One-Time Passwords (OTPs), fingerprint, and facial recognition authenticate users only during login and assume that the same user continues the session. This creates security risks if an authenticated device is stolen or accessed by an unauthorized individual. The objective of this project is to develop a behavioral biometric-based continuous authentication system that verifies user identity throughout an active banking session while ensuring a smooth user experience.

Recent research has identified behavioral biometrics as an effective approach for continuous authentication by analyzing user interactions such as touch behavior, swipe gestures, and smartphone sensor data [1], and keystroke dynamics [2, 4]. Machine learning techniques including One-Class Adversarial Autoencoders [3, 5] and Siamese Neural Networks [2] have demonstrated promising results in user authentication. However, many existing systems rely on a single behavioral modality, require large labeled datasets, or fail to adapt to changes in user behavior over time.

The proposed system employs a multimodal behavioral biometric framework that combines typing patterns, touch interactions, swipe gestures, and device motion to construct individualized user profiles. Anomaly detection algorithms such as One-Class SVM, Isolation Forest, and Autoencoders will be comparatively evaluated to identify deviations from legitimate user behavior. A dynamic trust score will be generated continuously to determine whether to maintain the current session, request secondary authentication, or terminate access when suspicious activity is detected. To demonstrate real-world feasibility, a mobile application will be developed to showcase the system's functionality in a live banking scenario, along with an API for seamless integration with third-party platforms.

The expected outcome is an intelligent, secure, and user-friendly continuous authentication framework for mobile banking applications. The proposed models will be evaluated using performance metrics such as False Acceptance Rate (FAR), False Rejection Rate (FRR), Equal Error Rate (EER), Precision, Recall, F1-Score, and authentication latency. The project aims to provide a practical and scalable solution for enhancing mobile banking security while contributing to research in behavioral biometrics.

Project Guide
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